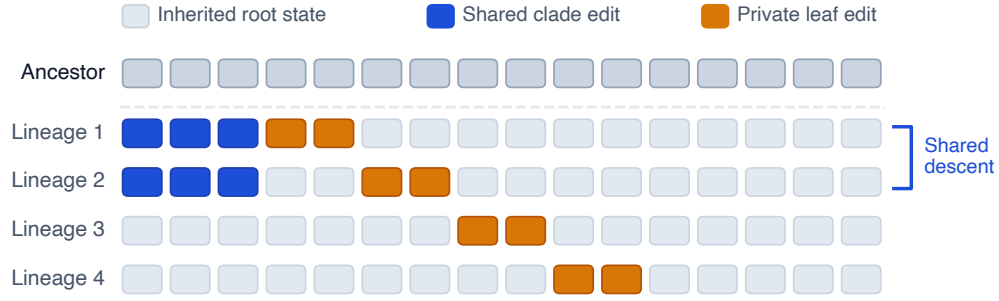
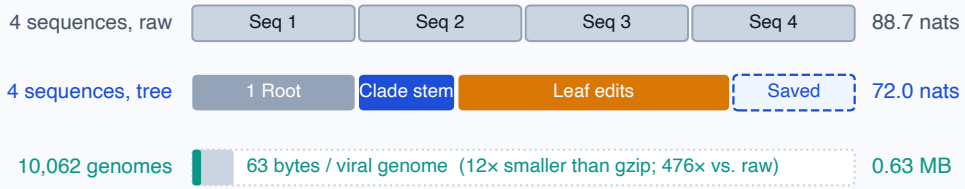
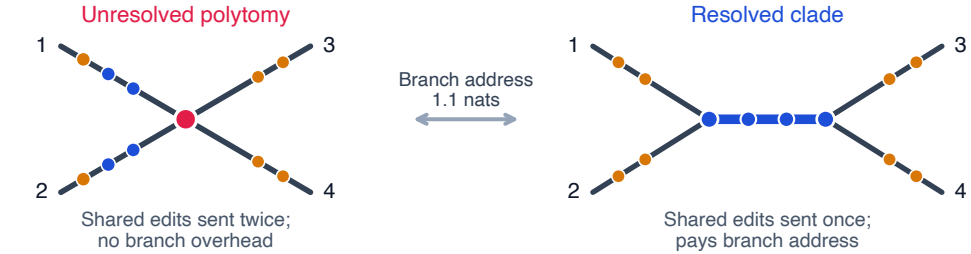
A Heredity as lossless compression



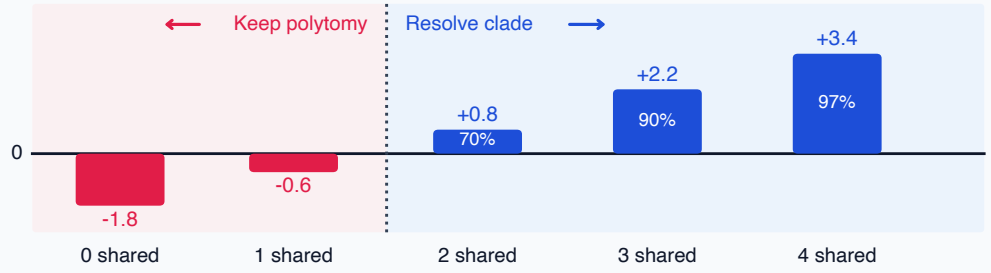
Physical description length required for exact lossless recovery



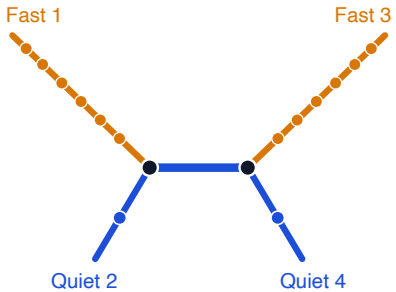
B When a branch earns its place: burst vs. split



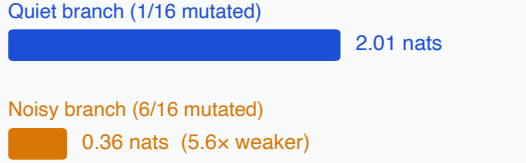
Net information saved by adding the branch (nats; % = posterior support)



C Uneven evolutionary rates and deep-time saturation

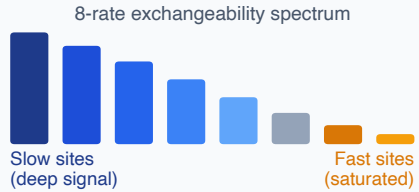


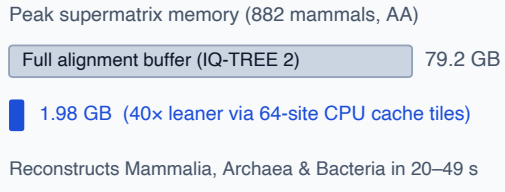
Positional information saved per shared match



Chance collisions on fast branches are naturally discounted

Deep-time supermatrices: site-rate spectrum & cache-tiled Fisher MDL

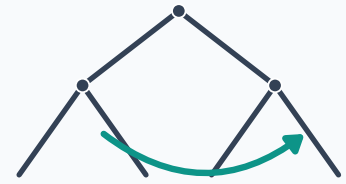




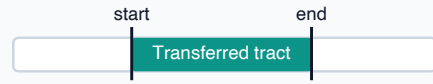
Reconstructs Mammalia, Archaea & Bacteria in 20–49 s

D Separating gene transfer from site convergence

Contiguous block transfer



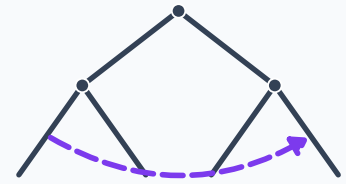
Lateral block transfer



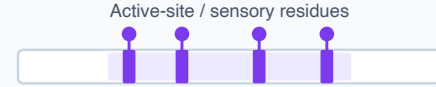
Link cost vs. shared edits explained

Recombination • HGT • Gene conversion

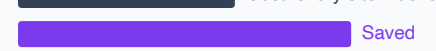
Scattered site convergence



Parallel convergence



Link cost vs. shared edits explained



Inner-ear hearing • Diet • Antibody CDRs